

- CareerOrbit V3.1 — Deployment & Infrastructure Manifest
 - 1. Infrastructure Architecture
 - Tiers
 - 2. Environment Configuration
 - 3. Database Management & Migrations
 - Essential Commands
 - 4. Build & CI/CD Pipeline
 - Build Lifecycle
 - Scaling & Performance Tuning
 - 5. Security & Data Isolation
 - 6. Deployment Checklist (For the Team)
 - 7. Disaster Recovery

CareerOrbit V3.1 — Deployment & Infrastructure Manifest

Audience: DevOps Engineers, Backend Leads, Project Owners **Objective:** A fail-safe manual for deploying, scaling, and maintaining the CareerOrbit production environment.

1. Infrastructure Architecture

CareerOrbit is a cloud-native application optimized for **Vercel** and **Neon PostgreSQL**.

Tiers

- **Compute:** Vercel (Next.js 15 Serverless & Edge Functions).
 - **Database:** Neon DB (Serverless PostgreSQL with horizontal autoscaling).
 - **Storage:** Metadata and cognitive states are stored in JSONB clusters.
 - **AI Inference:** OpenAI API (GPT-5.4/GPT-4o-mini).
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2. Environment Configuration

The following variables must be configured in Vercel's environment settings for the production deployment.

Variable Name	Description	Example/Format
DATABASE_URL	Primary Neon DB connection string.	postgres://user:pass@ep-host.neon.tech/neondb
OPENAI_API_KEY	Key for GPT-5/GPT-4 inference.	sk-proj-...
NEXT_PUBLIC_APP_URL	Base URL of the live app.	https://career-orbit.vercel.app

3. Database Management & Migrations

We use **Drizzle Kit** to manage schema changes. Unlike traditional migrations, we use the "push" workflow for rapid serverless iteration.

Essential Commands

- **Sync Schema (Dev/Staging):** `npx drizzle-kit push` *This pushes local `schema.ts` changes directly to the live Neon instance.*
- **Inspect Database:** `npx drizzle-kit studio` *Launches a local GUI to view live production data (use with caution).*

4. Build & CI/CD Pipeline

Vercel automatically handles the build process upon pushing to the `main` branch.

Build Lifecycle

1. **Pre-build:** Runs `npm install`.
2. **Build:** Runs `npm run build` (`next build`).
 - Executes static analysis and route optimization.

- Verifies Drizzle schema types.

3. **Post-build:** Deploys artifacts to Vercel's global CDN.

Scaling & Performance Tuning

- **Connection Pooling:** The `@neondatabase/serverless` driver handles pool management automatically. Ensure the `DATABASE_URL` uses the pooled endpoint (usually port 5432).
 - **Edge Runtime:** Critical routes (like the AI streaming endpoints) are optimized for the **Edge Runtime** to minimize TTFB (Time to First Byte).
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5. Security & Data Isolation

1. **SSL/TLS:** Enforced by Vercel; all HTTP traffic is automatically upgraded to HTTPS.
 2. **CSRF Protection:** Managed by NextAuth.js middleware.
 3. **Database Isolation:** Every user has a unique `userId` in the `users` table. All queries in `actions.ts` MUST include a `.where(eq(schema.table.userId, currentUserId))` clause to prevent data leakage.
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6. Deployment Checklist (For the Team)

Before every production release, verify the following:

- ☐ Run `npm run lint` to ensure no syntax errors.
 - ☐ Run `npm run test` (Vitest) to verify BKT math and AI schemas.
 - ☐ Ensure all new environment variables are added to Vercel dashboard.
 - ☐ Run `npx drizzle-kit push` to sync any schema changes.
 - ☐ Check Vercel logs for any "Function Timeout" warnings on the roadmap generation route.
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7. Disaster Recovery

- **Database Backups:** Neon DB automatically performs daily snapshots with a 7-day PITR (Point-in-Time Recovery) window.
- **Rollbacks:** Vercel allows instant rollbacks to any previous successful deployment with a single click in the dashboard.